

04

Creep Failure

Comprehensive Study Notes

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CREEP FAILURE (High Temperature Failures)

Creep is the term used to describe the tendency of a material to deform permanently in service. Creep is more severe in materials that are subjected to heat for long periods and near melting point. [Creep is not necessarily a failure mode, but is instead a damage mechanism]

Material deformation occurs as a result of long-term exposure to levels of stress that are below the yield or ultimate strength of the material.

The rate of this damage is a function of:

- Material Properties
- Exposure Time
- Exposure Temperature
- Applied Load (Stress): Depending on the magnitude of the applied stress and its duration, the deformation may become so large that a component can no longer perform its function.

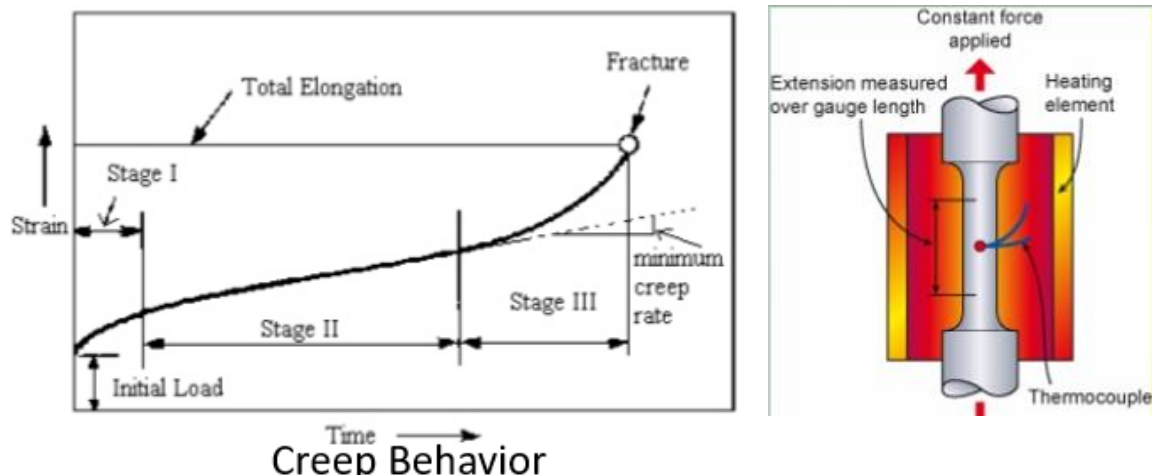
EX/ Creep of a turbine blade will cause the blade to contact the casing, resulting in the failure of the blade.

Characteristics:

- Creep failure is essentially plastic deformation, so its appearance is often similar to that of Ductile Fracture.
- Creep would be suspected if a component shows extensive plastic deformation before failure and if the material is operating at a temperature within its creep range, which is usually about **0.3 of its melting point** (in degrees K). [As with fatigue, the final failure may be ductile (by growth of voids) or brittle (by crack propagation)]

Creep Test:

Carried out by applying a constant load to a tensile specimen maintained at a constant temperature.



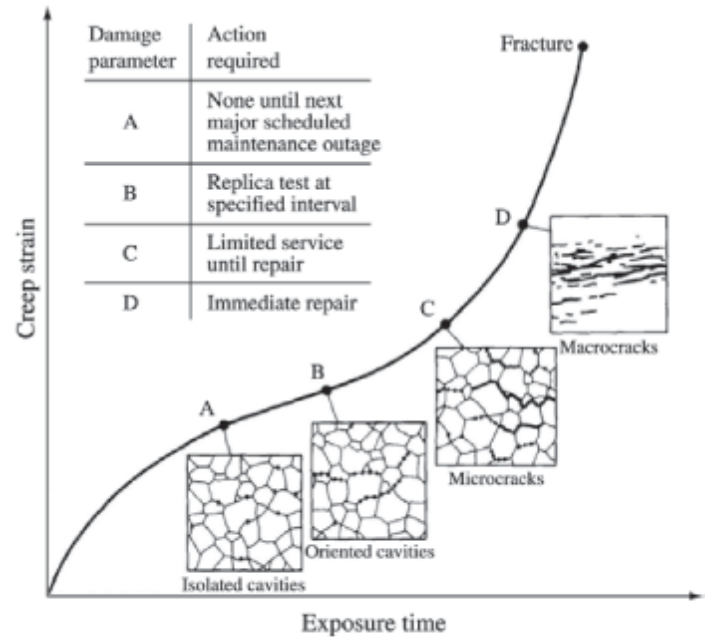
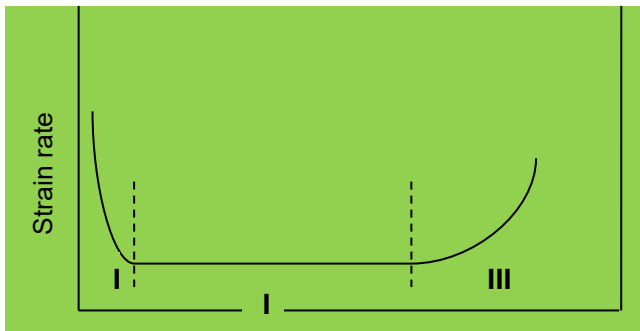
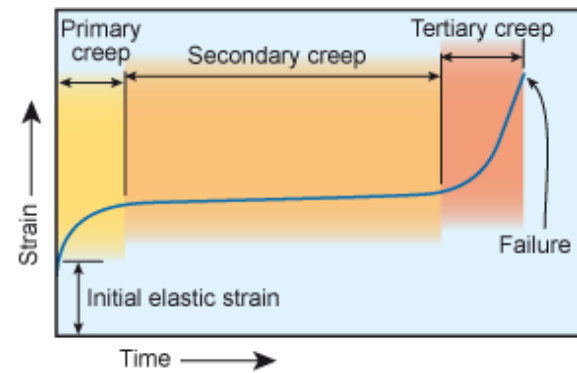


Figure 1. Creep life assessment based on cavity classification².

1. Stage I: Primary Creep
 - Period of decreasing creep rate.
 - Primary creep is a period of primarily transient creep.
 - During this period deformation takes place and the resistance to creep increases until stage II. (Strain Hardening)
2. Stage II: Secondary Creep
 - Period of roughly constant creep rate. Where **strain hardening = strain softening**.
 - Where steady state deformation occurs. **Dislocation annihilation, GBS, vacancy diffusion**, etc.
3. Stage II: Tertiary Creep
 - Occurs when there is a reduction in cross sectional area due to necking or effective reduction in area due to internal void formation. (Strain Softening)
 - Grain coarsening
 - Dislocation density decreases
 - Overaging of particles and precipitates
 - Void nucleation, growth, and coalescence

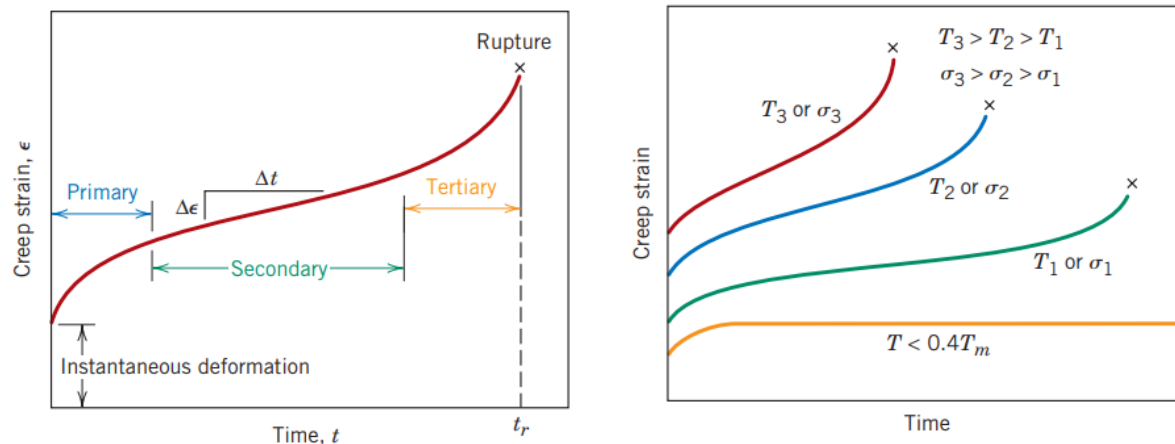
Constitutive Equations:

- Strain hardening: $\sigma = K\epsilon^n$
- Strain hardening & softening: $\sigma = K\epsilon^n\dot{\epsilon}^m$

Where $\dot{\epsilon}$ is strain rate and m strain rate sensitivity. Strain rate sensitivity is a measure of the sensitivity of the material to strain rate changes.

- Grain Boundary Sliding (GBS): $\sigma = K\dot{\epsilon}^m$

Creep Behavior:



A typical creep test consists of subjecting a specimen to a constant load or stress while maintaining the temperature constant; deformation or strain is measured and plotted as a function of elapsed time. Most tests are the constant load type, which yield information of an engineering nature. Upon application of the load there is an instantaneous deformation, as indicated in the figure, which is mostly elastic. The resulting creep curve consists of three regions, each of which has its own distinctive strain–time feature.

- Primary or transient creep occurs first, typified by a continuously decreasing creep rate; that is, the slope of the curve diminishes with time. This suggests that the material is experiencing an increase in creep resistance or strain hardening—deformation becomes more difficult as the material is strained.
- Secondary creep, sometimes termed steady-state creep, the rate is constant; that is, the plot becomes linear. This is often the stage of creep that is of the longest duration. The constancy of creep rate is explained on the basis of a balance between the competing processes of strain hardening and recovery, recovery being the process whereby a material becomes softer and retains its ability to experience deformation. The minimum or steady-state creep rate is the slope of the linear segment in the secondary region. It is the engineering design parameter that is considered for long-life applications.
- In tertiary creep, there is an acceleration of the rate and ultimate failure. This failure is frequently termed rupture and results from microstructural and/or metallurgical changes,

for example, grain boundary separation, and the formation of internal cracks, cavities, and voids. Also, for tensile loads, a neck may form at some point within the deformation region. These all lead to a decrease in the effective cross-sectional area and an increase in strain rate.

Both temperature and the level of the applied stress influence the creep characteristics. At a temperature substantially below $0.4T_m$ and after the initial deformation, the strain is virtually independent of time. With either increasing stress or temperature, the following will be noted:

1. The instantaneous strain at the time of stress application increases
2. The steady-state creep rate is increased
3. The rupture lifetime is diminished

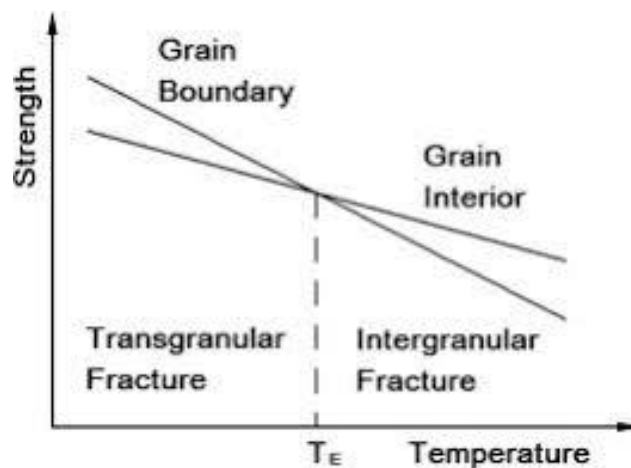
Creep Failure Mechanism

Creep failure is essentially plastic deformation, so its appearance is often similar to that of Ductile Fracture. Creep would be suspected if a component shows extensive plastic deformation before failure and if the material is operating at a temperature within its creep range.

Microscopic Appearance: Creep failures are usually of the intergranular ductile type, due to the formation of voids on the grain boundaries. Failure is usually preceded by a lot of microscopic damage, which can be seen on polished sections. This damage takes the form of voids and/or small cracks, mostly located on the grain boundaries.

Creep depends on operating temperature, which is either higher or lower than the Equi-cohesive temperature. Equi-cohesive temperature is the temperature that corresponds to about $0.7-0.8 T_m$ of the material.

- AT Temperature Below Equi-cohesive temperature, usually failure occurs by Vacancy Creep.
- At higher temperatures with other conditions creep failure occurs with GBS.



The conditions for creep failure to occur with Grain Boundary Sliding (GBS) at higher temperatures include:

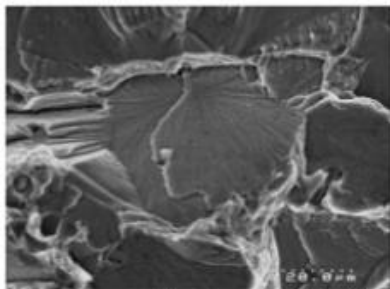
- High temperature: Creep failure typically occurs at temperatures significantly above the material's recrystallization temperature. This temperature is usually higher than the Equi-cohesive temperature.
- Applied stress: Creep failure is influenced by the magnitude of the applied stress. It occurs under relatively low stresses compared to the material's ultimate tensile strength. The stress can be applied either continuously or intermittently.
- Time: Creep failure is time-dependent and occurs over an extended period. It involves the slow and progressive deformation of the material under constant stress.
- Grain boundary sliding: In GBS, the primary mechanism of deformation is the relative sliding of individual grains along their boundaries. This sliding is facilitated by the presence of grain boundary films, impurities, or other defects. GBS allows for plastic deformation to occur at lower stress levels than would be required for dislocation motion.
- Grain boundary diffusion: Diffusion processes at the grain boundaries, such as grain boundary self-diffusion or diffusion of impurity atoms along the boundaries, can contribute to creep failure by facilitating grain boundary sliding.
- Intergranular void formation: As the material undergoes creep, voids can form along the grain boundaries due to a variety of mechanisms, including diffusion and grain boundary sliding. These voids contribute to the eventual failure of the material.

INCREASING TEMPERATURE

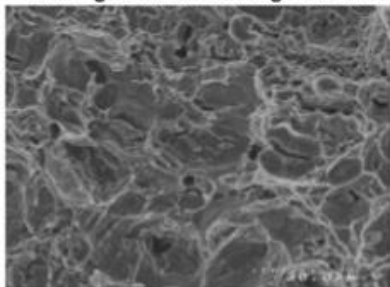
[At T just below $T_{\text{Recrystallization}}$, Ductility drops due to grain boundary sliding (Intergranular Failure)]

Transgranular Fracture

Slip Planes are Weaker than Grain Boundaries



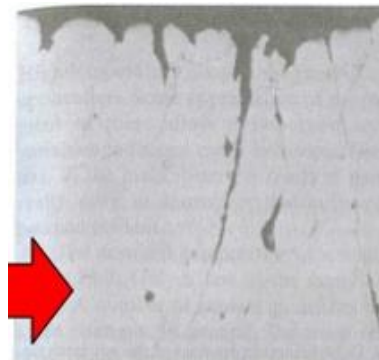
Transgranular cleavage fracture



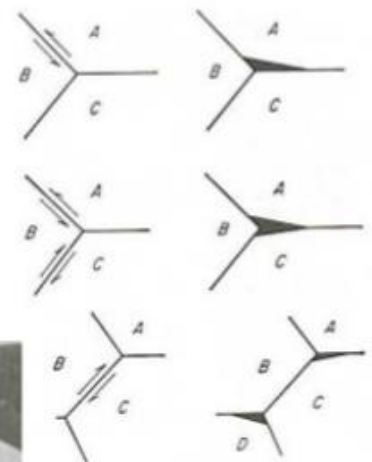
Transgranular microvoid coalescence

Intergranular Fracture

Grain Boundaries are Weaker than Slip Planes



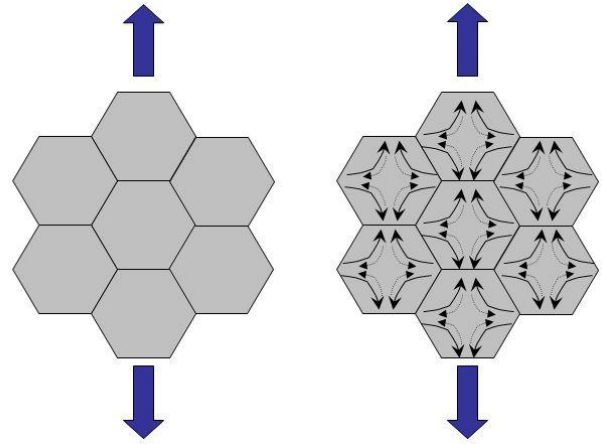
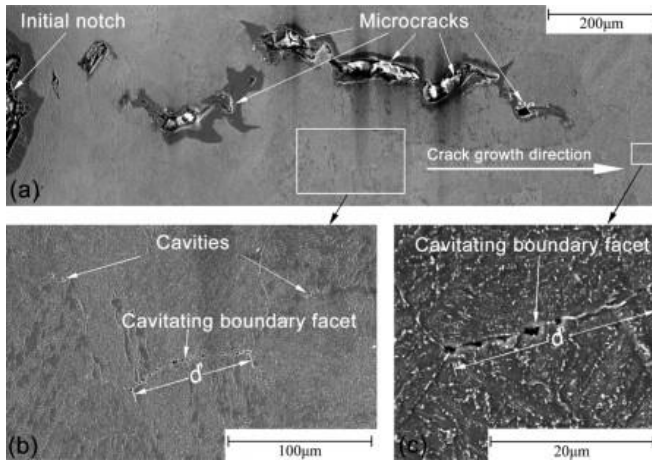
Grain boundary fracture



The formation of intergranular crack by grain boundary sliding

1. Creep Failure by Vacancy Creep

Vacancy Diffusion during Creep when temperature is lower than the Equicohesive temperature ($0.7-0.8T_m$).



2. Creep failure by Grain Boundary Sliding:

Requirements:

- Operating temperature $>$ Equicohesive temperature
- Fine Equiaxed Grains $< 10\mu m$
- Very low Strain Rate

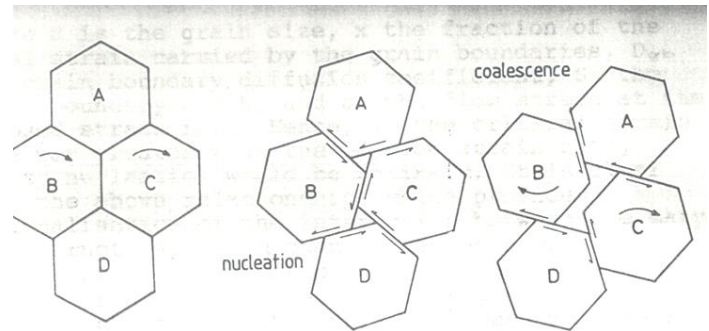
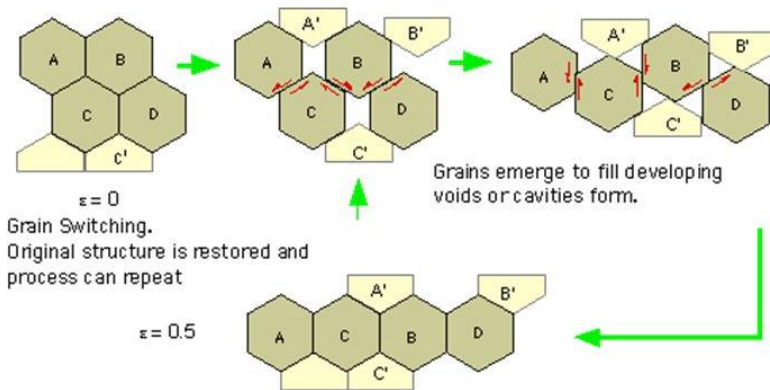
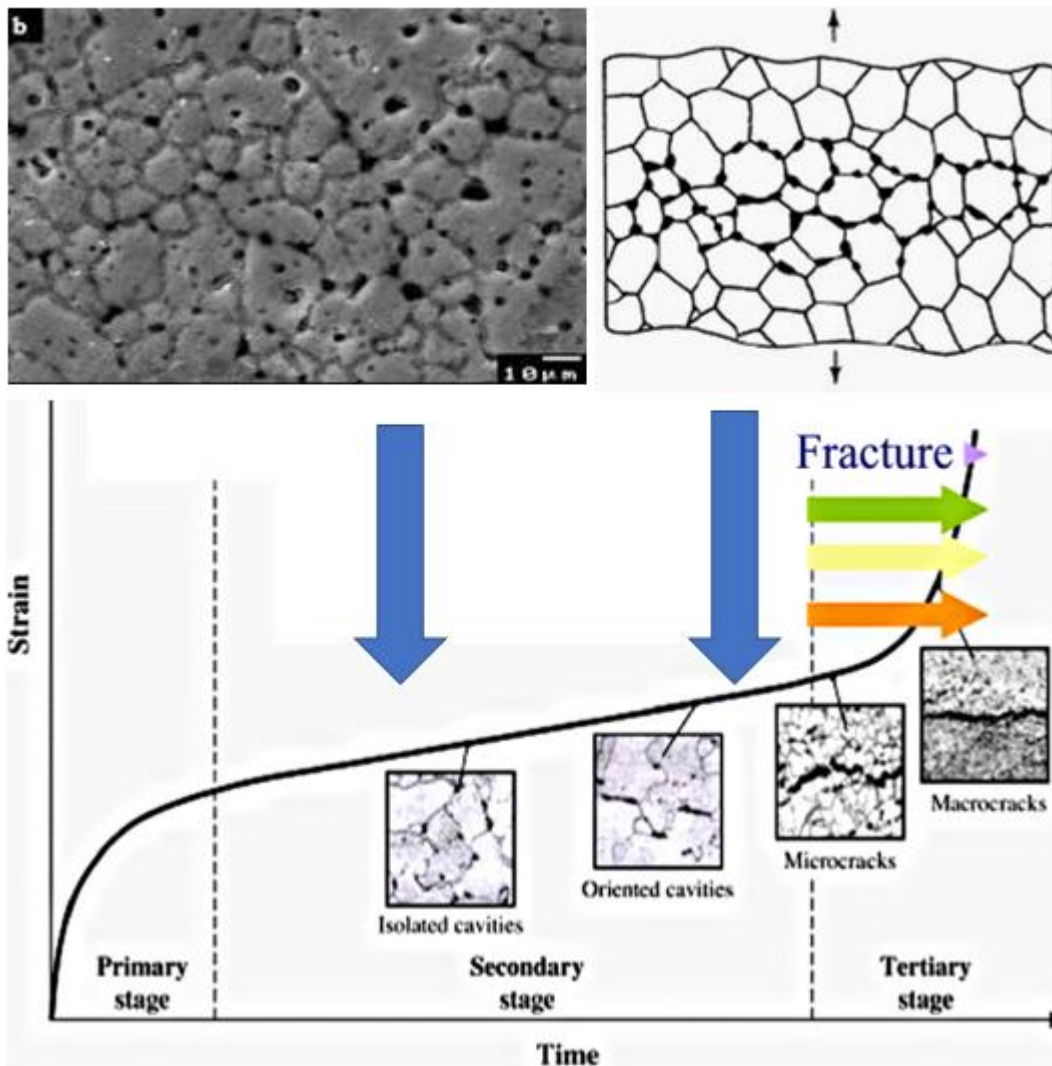
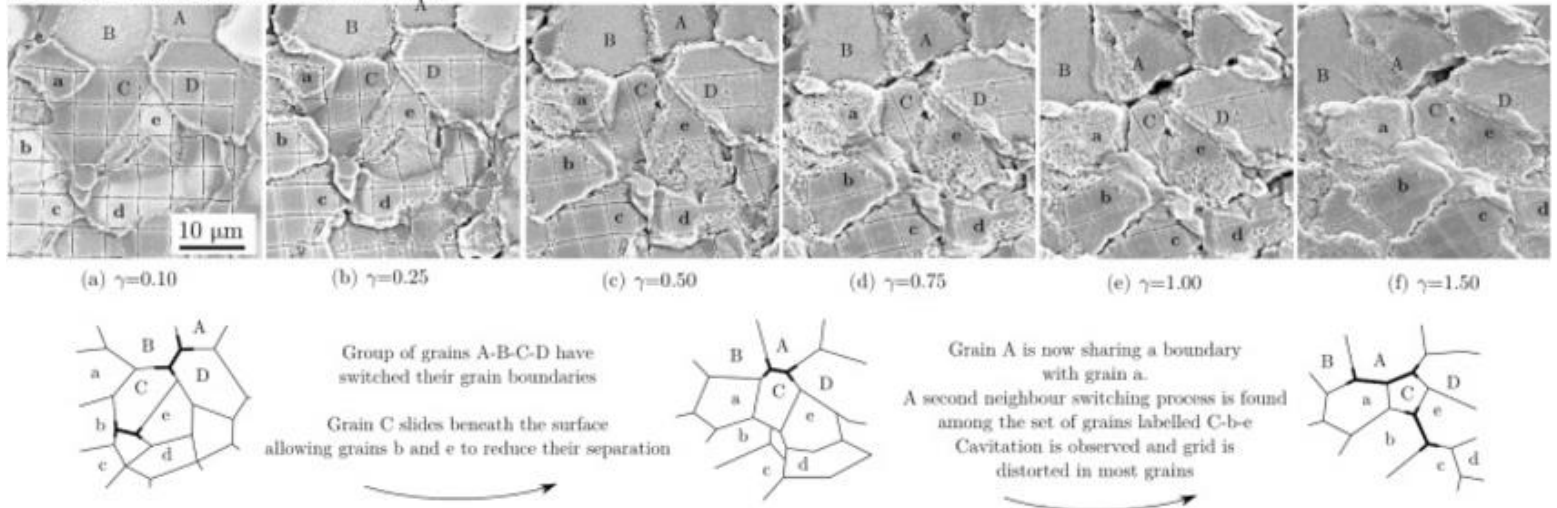
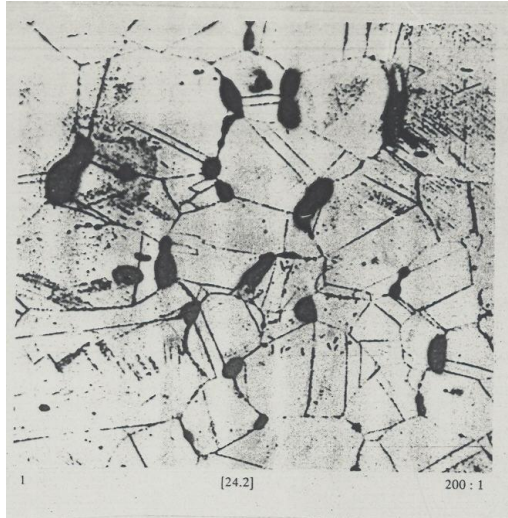


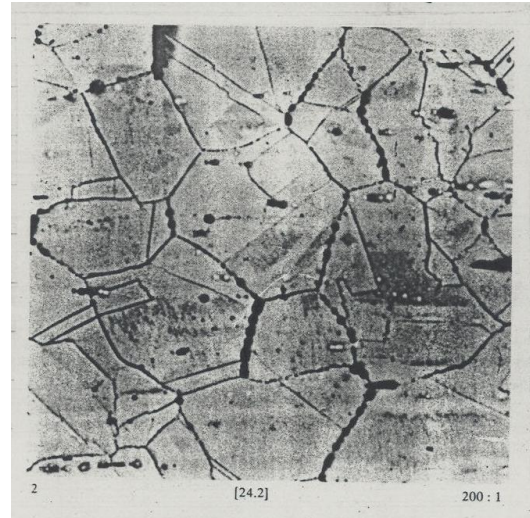
Figure 5.15 Schematic illustration of void nucleation as a result of unaccommodated grain rotation.

Detail B: Multiple grain neighbour switching





GBS



Vacancy Creep

Creep Failure by Vacancy Creep (At temperatures below Equi-cohesive temperature → Transgranular Fracture):

Microstructure features:

- Initial notch or stress concentration: A localized stress concentration or notch in the specimen can initiate the creep process.
- Transgranular fracture: Vacancy creep primarily leads to transgranular fracture, meaning that cracks propagate through the grains themselves.
- Void formation: Under creep conditions, vacancies or atomic defects accumulate within the crystal lattice.
- Cavities: The accumulation of vacancies can lead to the formation of cavities or voids within the material.

Microstructural fracture mechanisms:

- Vacancy diffusion and void growth: Vacancies diffuse through the lattice and cluster together, leading to void formation.
- Void coalescence: The voids can grow and coalesce with neighboring voids, resulting in the formation of larger voids.
- Transgranular fracture: As voids grow and coalesce, they act as stress concentrators and promote crack initiation and propagation through the grains.

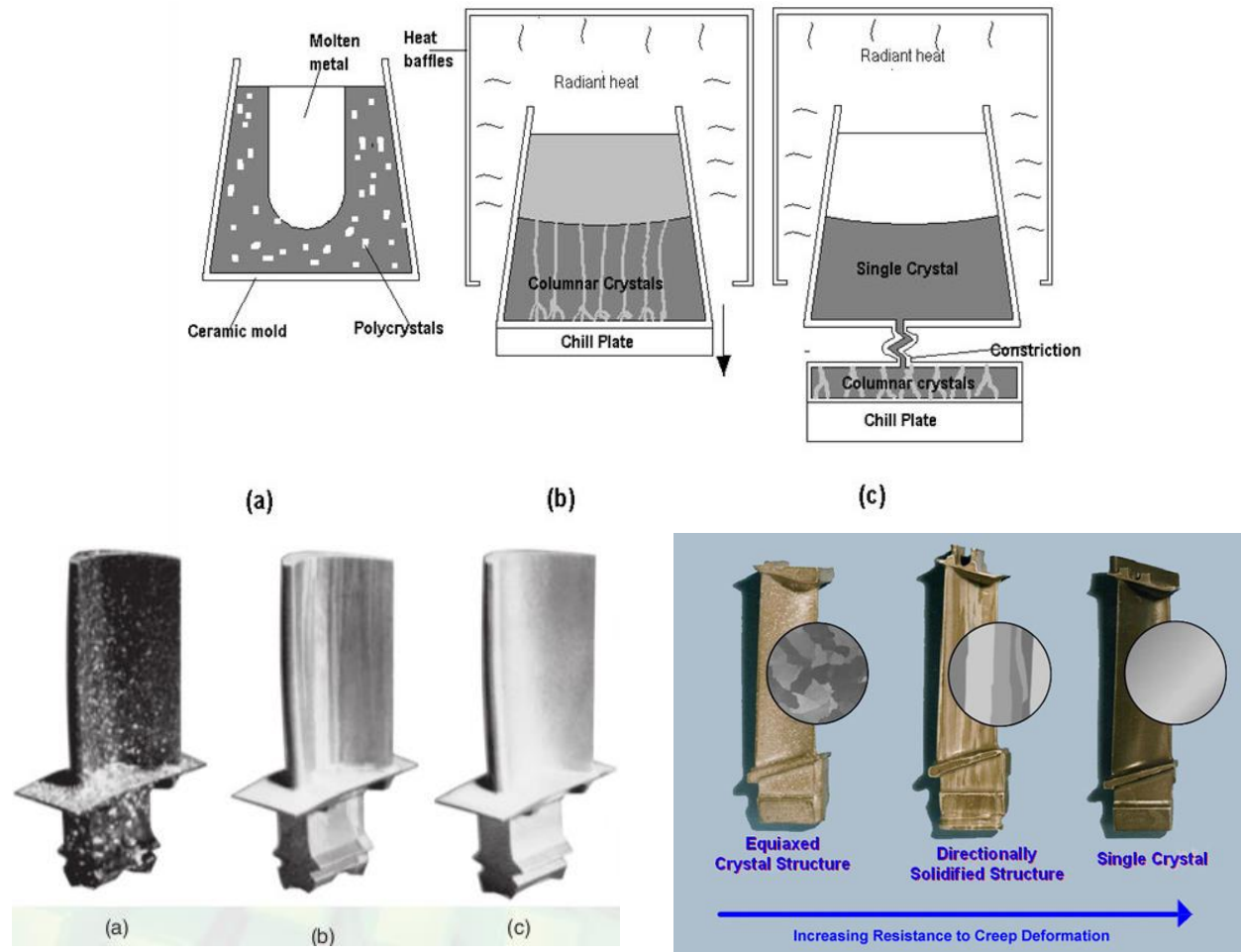
Creep Failure by Grain Boundary Sliding (At higher temperatures with other conditions → Intergranular Fracture):**Microstructure features:**

- High temperature: Creep failure by Grain Boundary Sliding occurs at higher temperatures, usually above the Equi-cohesive temperature.
- Grain switching: At lower strain rates, grain boundaries can slide relative to each other, allowing the material to deform with minimal microstructural changes.
- Nucleation of grains: At higher strain rates, new grains can nucleate within the material to fill developing voids or cavities.
- Intergranular fracture: The deformation and stress concentration at grain boundaries can lead to intergranular fracture.

Microstructural fracture mechanisms:

- Grain boundary sliding: The primary deformation mechanism is the sliding of grain boundaries, which allows for plastic deformation without significant dislocation movement.
- Grain boundary cavitation: Stress concentrations along grain boundaries can lead to the formation of cavities or voids.
- Intergranular fracture: The accumulation of damage, such as cavities or voids, along the grain boundaries can result in intergranular fracture, where cracks propagate along the boundaries.

Turbine Blade Casting



- a) Conventional casting with ceramic mold: Use of complicated alloys with alloying elements suitable for structural stability at elevated temperatures.
- b) Directional Solidification: Shows columnar grains represented by vertical streaks.
- c) Single crystal: Have properties at high temperatures superior to other blades.

SUMMARY

Creep is the change in dimension of a metal or alloy under an applied stress. It occurs at a temperature generally above the recrystallization temperature. Creep strain may cause sufficiently large changes in the dimensions of a component to render it useless for further service. The creep strain may lead to fracture. This type of failure is called stress rupture. Creep properties become important in the following areas:

- The soft metals are used at about room temperature such as lead pipes and white metal bearings.
- Steam and chemical plants operating at 450-550°C.
- Gas turbines working at high temperatures.

Characteristics:

- Easy to recognize by the local deformation and multiplicity of intergranular cracks that are usually present adjacent to the main crack (Fig. 4.28).
- Section close to the fracture shows at higher magnification multiplicity of creep voids at grain boundaries (Fig. 4.29).
- Cracks originate at triple points.

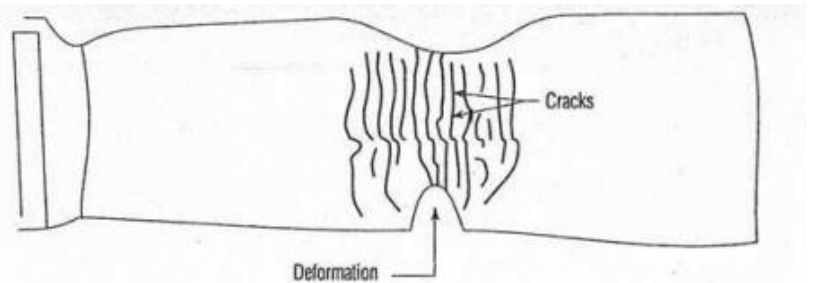


Fig. 4.28 Schematic representation of creep deformation and cracks in a turbine blade



Fig. 4.29 Creep voids at the grain boundaries and cracking at triple points in the microstructure of a creep failed Cu-Cr material ($\times 400$)